Module IV: Theorems, Figures, Cross References and Presentation

Study Notes · MAT5FS112 · Syllabus units 12–19

Abstract. This note covers the AMS theorem package, boxes, floating figures, cross references, footnotes and the layout conventions of a presentable document. Each example compiles as it stands.

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Figure 1: A curve from the origin, with its chord drawn dashed.

1 Theorems with `amsthm`

1.1 Defining and using a theorem

The preamble defines environments with `\newtheorem{name}{Title}`. In the body the environment is used like any other:

```
\newtheorem{theorem}{Theorem}[section]       % numbered within sections
\newtheorem{lemma}[theorem]{Lemma}           % shares the theorem counter
\newtheorem{definition}[theorem]{Definition}
\newtheorem{remark}[theorem]{Remark}

\begin{theorem}[Pythagoras]\label{thm:pyth}
In a right-angled triangle with legs $a, b$ and hypotenuse $c$,
$a^2 + b^2 = c^2$.
\end{theorem}
```

Theorem 1.1 (Pythagoras)

In a right-angled triangle with legs a, b and hypotenuse c, $a^2 + b^2 = c^2$.

Lemma 1.2

Every theorem in the same counter as Theorem 1.1 is numbered in sequence with it.

Definition 1.3

A triangle is *right-angled* if one of its angles equals 90°.

Remark 1.4

The theorem numbers include the section number because of the optional argument `[section]` in `\newtheorem`.

```
\begin{proof}
Let the right angle be at $C$. Place a square of side $a+b$ around the
triangle; comparing areas gives the result. \qedhere
\end{proof}
```

Proof.

Let the right angle be at C. Place a square of side $a + b$ around the triangle; comparing areas gives the result. ∎

1.2 Choosing a theorem style

```
\theoremstyle{plain}        % theorem, lemma, proposition, corollary
\theoremstyle{definition}   % definition, example, exercise
\theoremstyle{remark}       % remark, note, notation
```

STYLE	HEADING	BODY TEXT	USED FOR
`plain`	**bold**	*italic*	theorem, lemma, proposition, corollary
`definition`	**bold**	upright	definition, example, exercise
`remark`	*italic*	upright	remark, note, notation

2 Boxes

`\fbox` draws a frame around its contents. `\framebox` takes an optional width and alignment. `\makebox` sets a fixed width without a frame. `\parbox` makes a fixed-width paragraph in a line.

```
\noindent
\fbox{Boxed text}\quad
\framebox[4cm][c]{Centred in 4\,cm}\quad
\makebox[3cm][r]{Right in 3\,cm}\\[1em]
\fbox{\parbox{0.7\textwidth}{A framed paragraph. The \texttt{parbox}
lets the text wrap inside a box of fixed width, for key results or
warnings.}}
```

Boxed text Centred in 4 cm Right in 3 cm

A framed paragraph. The `parbox` lets the text wrap inside a box of fixed width, for key results or warnings.

3 Floating figures

A figure environment places the picture at a suitable location and numbers it. Here the picture is drawn with TikZ so that the example needs no image file. With an external picture, use `\includegraphics[width=0.6\textwidth]{file.png}` (`graphicx`).

```
\begin{figure}[htbp]
  \centering
  \begin{tikzpicture}[scale=1.2]
    \draw[->] (-0.3,0) -- (3.3,0) node[right] {$x$};
    \draw[->] (0,-0.3) -- (0,2.6) node[above] {$y$};
    \draw[thick, blue] (0,0) .. controls (1,2.2) and (2,2.2) .. (3,0.4);
    \draw[dashed] (0,0) -- (3,0.4);
    \node[below right] at (3,0.4) {end};
  \end{tikzpicture}
  \caption{A curve from the origin, with its chord drawn dashed.}
  \label{fig:curve}
\end{figure}
```

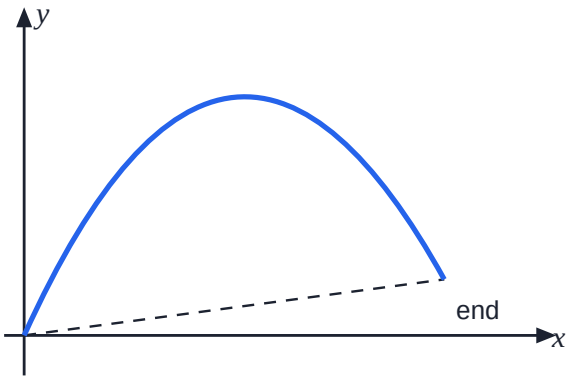


Figure 1. A curve from the origin, with its chord drawn dashed.

Figure 1 shows the curve. The caption sits below the figure; `\caption` before `\label` is required so that the reference shows the figure number.

4 Cross references

4.1 Labels and the three reference commands

A label is placed right after the item it names, inside its environment or heading. Then:

- `\ref{label}` gives the number: Section 1.
- `\pageref{label}` gives the page: page 1.
- `\eqref{label}` gives the equation number in brackets (for equations): (1).

```
\begin{equation}\label{eq:mass}
  F = ma
\end{equation}
```

$$F = ma \tag{1}$$

Equation (1) above is referred to by `\eqref{eq:mass}`.

4.2 Smart references with `hyperref`

With `hyperref` loaded, `\autoref{label}` writes the word for you and makes the reference a clickable link: see Figure 1 and Theorem 1.1.

4.3 Rules for labels

- Labels are case-sensitive and must be unique in the document.
- Prefixes such as `sec:`, `fig:`, `tab:`, `eq:`, `thm:` keep them readable.
- After adding or moving a label, compile again (twice in total) so that the references are updated.

> **Compile twice.** `\ref`, `\pageref` and `\listoffigures` need a second pass before they show the right numbers.

5 Footnotes

A footnote is written with `\footnote{...}`, which numbers it and places the text at the bottom of the page.[1] Inside a table or a `minipage`, use `\footnotemark` in the text and `\footnotetext` after the table.

[1] The footnote is typeset at the bottom of the page where it appears, not at the end of the document.

6 Presentation conventions

6.1 Front matter and back matter

A presentable document has, in order: a title block, an abstract, a table of contents, lists of figures and tables, the body, and then references and appendices. The sectioning commands fill the body. `\appendix` changes the later sections to letters.

6.2 Spacing and emphasis

- Put one blank line between paragraphs; do not use `\\` for that purpose.
- Use `\emph` for the first use of a term and `\textbf` sparingly.

- Keep display maths and figures close to the text that discusses them.

A A short appendix

Appendix sections are labelled A, B, …. Here the theorem counter continues to use the section number from the body, as Theorem 1.1 shows.